

Vectors in the plane and transformations

[Cambridge revision]

A vector is a journey, not a place — and every transformation is one rule applied to all of them.

LESSON 59–60

2 × 40 MIN

REVISION OF GRADE 8–9 VECTORS

IGX 23.1–23.3

LESSON CLOCK

20'

25'

20'

25'

10'

Vectors as journeys	20 min
Arithmetic of vectors	25 min
Proving with vectors	20 min
The four transformations	25 min
Homework	10 min

LEARNING OBJECTIVES

- Add, subtract and scale vectors in column form and geometrically.
- Compute the magnitude of a vector and the position vector of a point.
- Use vectors to prove that lines are parallel or that points are collinear.
- Describe fully a translation, reflection, rotation and enlargement.

TEXTBOOK

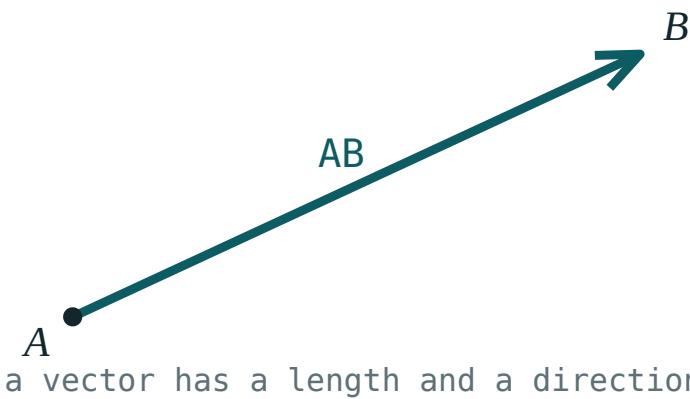
National	Geometry 9, pp. 147–170
Cambridge	Core & Extended, pp. 590–612
Workbook	Exercise 23.1–23.3

01

Explanation

Vectors as journeys

A vector has **magnitude** and **direction** but no position. The arrow from $(1, 1)$ to $(4, 5)$ and the arrow from $(0, 0)$ to $(3, 4)$ are the **same vector** $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$.



One vector, drawn twice — same length, same direction, different place.

$$|\begin{pmatrix} a \\ b \end{pmatrix}| = \sqrt{a^2 + b^2}$$

THE POSITION VECTOR FIXES IT IN PLACE

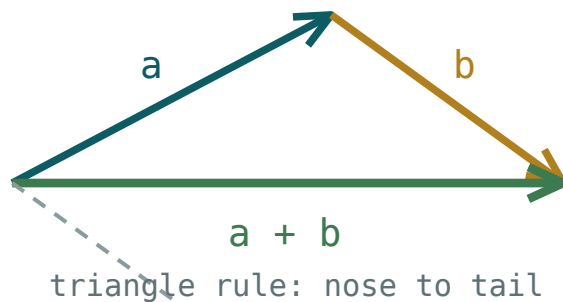
The position vector of A is OA, drawn from the origin. Then for any two points

$$AB = OB - OA \text{ (“to minus from”)}$$

Getting this the wrong way round reverses every vector in the question, so learn it as a sentence.

Arithmetic of vectors

OPERATION	IN COLUMNS	GEOMETRICALLY
add	$\begin{pmatrix} a \\ b \end{pmatrix} + \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} a+c \\ b+d \end{pmatrix}$	nose to tail
subtract	$\begin{pmatrix} a \\ b \end{pmatrix} - \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} a-c \\ b-d \end{pmatrix}$	add the reverse
scale	$k\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} ka \\ kb \end{pmatrix}$	same line, k times as long



Nose to tail — the resultant closes the triangle.

PARALLEL MEANS A SCALAR MULTIPLE

u and v are parallel exactly when $v = ku$ for some number k . So $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$ and $\begin{pmatrix} 9 \\ 12 \end{pmatrix}$ are parallel ($k = 3$); $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$ and $\begin{pmatrix} 4 \\ 3 \end{pmatrix}$ are not.

Unit vector. Divide by the magnitude: the unit vector along $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$ is $\frac{1}{5} \begin{pmatrix} 3 \\ 4 \end{pmatrix}$.

Proving with vectors

Two standard proofs cover most questions.

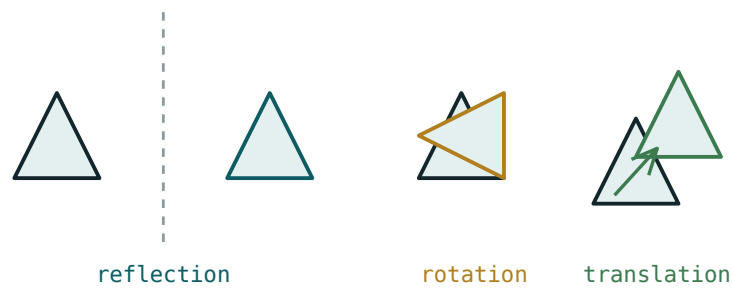
TO PROVE	SHOW
AB is parallel to CD	$AB = k \cdot CD$
A, B, C are collinear	$AB = k \cdot BC$
$ABCD$ is a parallelogram	$AB = DC$
M is the midpoint of AB	$OM = \frac{1}{2}(OA + OB)$

The collinearity test differs from the parallel test only in that the two vectors share a **point**. Parallel plus a common point means one line — that is the whole argument, and it should be written in the answer.

A VECTOR PROOF NEEDS A CONCLUDING SENTENCE

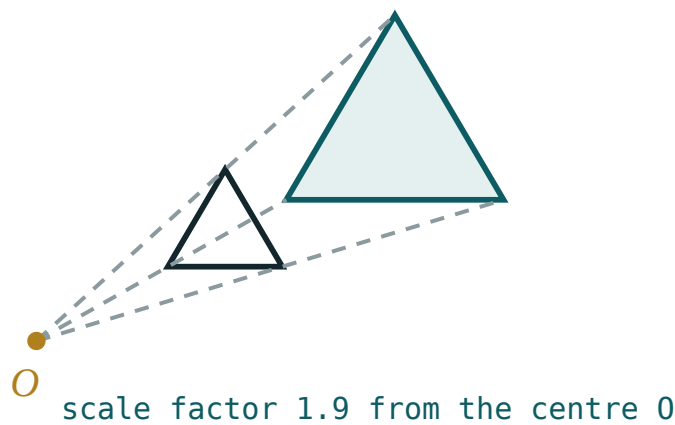
“ $AB = 2 \cdot BC$, and B is common to both, so A , B and C are collinear.” Without the second half the algebra proves only parallelism.

The four transformations



Reflection, rotation and translation — each carries the shape to a congruent copy.

TRANSFORMATION	FULLY DESCRIBED BY	PRESERVES
translation	a column vector	size, shape, direction
reflection	the mirror line	size and shape; reverses sense
rotation	centre, angle and sense	size and shape
enlargement	centre and scale factor k	shape only



An enlargement from a centre — lengths scale by k , areas by k^2 .

“FULLY DESCRIBE” IS A MARK SCHEME INSTRUCTION

A rotation needs **three** pieces of information; giving two earns part marks. An enlargement needs the centre as well as k . And a negative k puts the image on the other side of the centre, inverted.

Under an enlargement of factor k : lengths $\times k$, areas $\times k^2$. Under the other three, lengths and areas are unchanged — they are **congruences**.

02 Worked examples

EXAMPLE 1 With $A(1, 2)$ and $B(5, 9)$, find AB and $|AB|$.

$$\textcircled{1} \quad AB = OB - OA$$

“to minus from”.

$$\textcircled{2} \quad = \begin{pmatrix} 5 \\ 9 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 4 \\ 7 \end{pmatrix}$$

$$\textcircled{3} \quad |AB| = \sqrt{16 + 49} = \sqrt{65} \approx 8.06$$

ANSWER

$$\begin{pmatrix} 4 \\ 7 \end{pmatrix}, \sqrt{65}$$

EXAMPLE 2 Show that $A(1, 1)$, $B(3, 4)$ and $C(7, 10)$ are collinear.

$$① \quad AB = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

$$② \quad BC = \begin{pmatrix} 4 \\ 6 \end{pmatrix}$$

$$③ \quad BC = 2 \cdot AB$$

Parallel.

$$④ \quad B \text{ lies on both, so the three are collinear.}$$

The concluding sentence.

ANSWER

Collinear, since $BC = 2 \cdot AB$ and B is common

EXAMPLE 3 Find the unit vector in the direction of $\begin{pmatrix} -6 \\ 8 \end{pmatrix}$.

$$① \quad |v| = \sqrt{36 + 64} = 10$$

$$② \quad \frac{1}{10} \begin{pmatrix} -6 \\ 8 \end{pmatrix}$$

$$③ \quad = \begin{pmatrix} -0.6 \\ 0.8 \end{pmatrix}$$

Check: $0.36 + 0.64 = 1$.

ANSWER

$$\begin{pmatrix} -0.6 \\ 0.8 \end{pmatrix}$$

EXAMPLE 4 A triangle of area 12 is enlarged by factor -2 about $(1, 1)$. Describe the image.

- ① Lengths double.

- ② Area $\times k^2 = 4$, so 48.
 k^2 is positive either way.

- ③ The image is on the opposite side of $(1, 1)$, inverted.

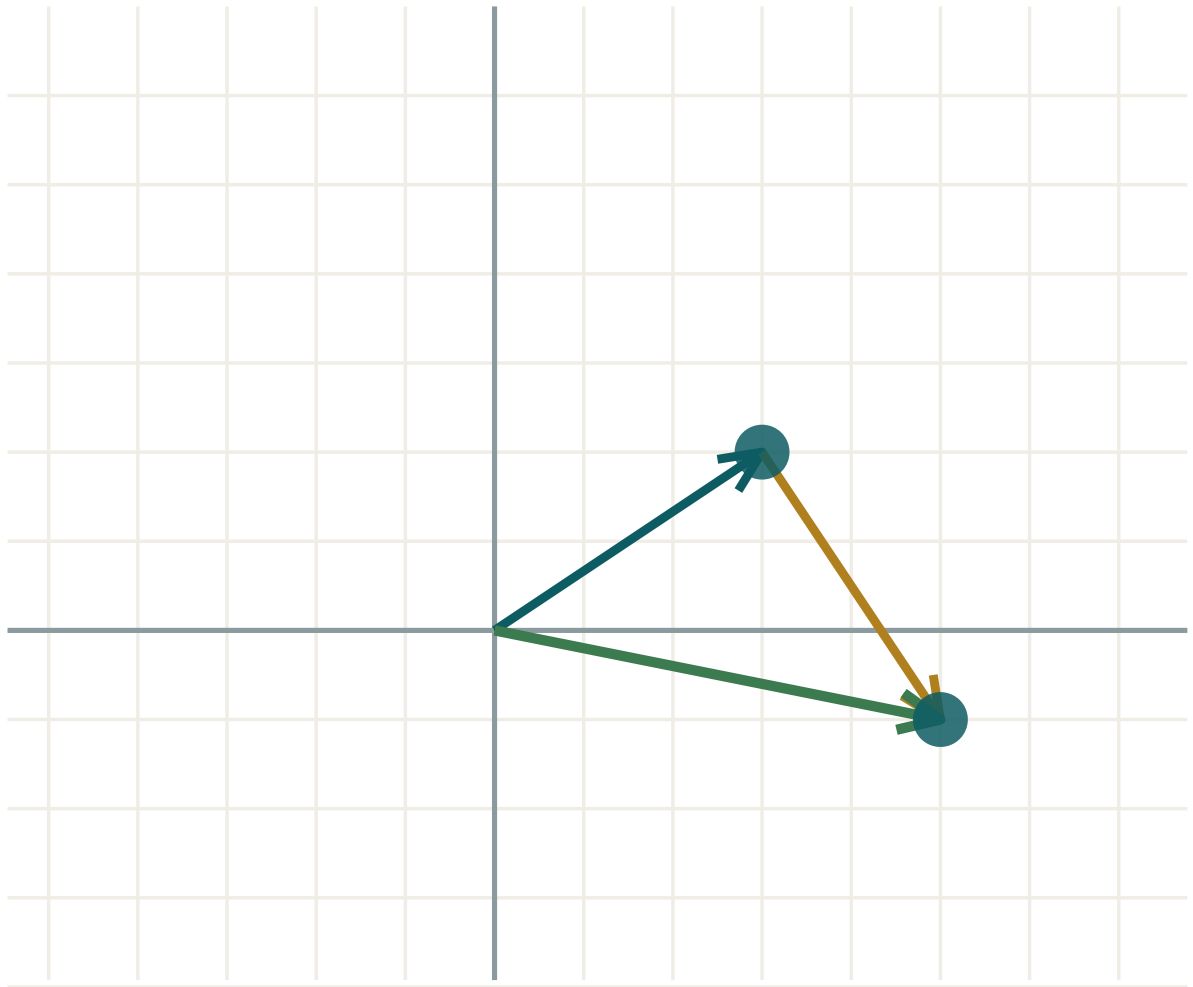
ANSWER

Area 48, inverted, on the far side of the centre

03 Interactive model

Cut a triangle from card and physically translate, reflect, rotate and enlarge it on squared paper.

Adding vectors nose to tail



$a \ (3; \ 2)$

$b \ (2; \ -3)$

$a + b \ (5; \ -1)$

$|a| \ 3.61$

$|b| \ 3.61$

$|a + b| \ 5.1$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Vector	Vektor	Вектор
Column vector	Ustun vektor	Вектор-столбец
Magnitude	Modul	Модуль
Position vector	Radius-vektor	Радиус-вектор
Resultant	Teng ta'sir etuvchi	Равнодействующая
Scalar multiple	Skalyarga ko'paytma	Умножение на число
Translation	Parallel ko'chirish	Параллельный перенос
Reflection	Simmetriya	Отражение
Rotation	Burish	Поворот
Enlargement	Gomotetiya	Гомотетия
Scale factor	O'xshashlik koeffitsiyenti	Коэффициент подобия
Invariant point	O'zgarmas nuqta	Неподвижная точка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

AB equals:

$|(3, 4)|$ is:

Two vectors are parallel when:

Collinearity needs, beyond parallel vectors:

equal length

a common point

the origin

nothing

A rotation is fully described by:

the angle

the centre and angle

centre, angle and sense

the centre

Under an enlargement of factor 3, areas multiply by:

3

6

9

27

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\begin{pmatrix} 2 \\ 3 \end{pmatrix} + \begin{pmatrix} 4 \\ 1 \end{pmatrix}$

ANSWER $\begin{pmatrix} 6 \\ 4 \end{pmatrix}$

2. $\begin{pmatrix} 5 \\ 2 \end{pmatrix} - \begin{pmatrix} 1 \\ 6 \end{pmatrix}$

ANSWER $\begin{pmatrix} 4 \\ -4 \end{pmatrix}$

3. $3\begin{pmatrix} 2 \\ -1 \end{pmatrix}$

ANSWER $\begin{pmatrix} 6 \\ -3 \end{pmatrix}$

4. $\left| \begin{pmatrix} 3 \\ 4 \end{pmatrix} \right|$

ANSWER 5

5. $\left| \begin{pmatrix} 5 \\ 12 \end{pmatrix} \right|$

ANSWER 13

6. AB for $A(1, 2)$, $B(5, 9)$

ANSWER $\begin{pmatrix} 4 \\ 7 \end{pmatrix}$

7. Area factor of an enlargement with $k = 4$

ANSWER 16

Medium · the standard question

7 PROBLEMS

1. $|AB|$ for $A(1, 2), B(5, 9)$

ANSWER $\sqrt{65} \approx 8.06$

2. Unit vector along $\begin{pmatrix} -6 \\ 8 \end{pmatrix}$

ANSWER $\begin{pmatrix} -0.6 \\ 0.8 \end{pmatrix}$

3. Are $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$ and $\begin{pmatrix} 9 \\ 12 \end{pmatrix}$ parallel?

ANSWER Yes, $k = 3$

4. Are $A(1,1), B(3,4), C(7,10)$ collinear?

ANSWER Yes — $BC = 2 \cdot AB$

5. Midpoint of $A(2, 3)$ and $B(8, 11)$ as a position vector

ANSWER $\begin{pmatrix} 5 \\ 7 \end{pmatrix}$

6. Describe fully: $(1,1) \rightarrow (4,3)$ for every point

ANSWER Translation by $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$

7. A shape of area 12 enlarged by $k = -2$: new area

ANSWER 48

Hard · stretch and reason

7 PROBLEMS

1. Find k so that $\begin{pmatrix} k \\ 6 \end{pmatrix}$ is parallel to $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$

ANSWER $k = 4$

2. $OP = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, $OQ = \begin{pmatrix} 7 \\ 5 \end{pmatrix}$: find the point dividing PQ in $1 : 2$

ANSWER $(3, 3)$

3. Prove that the diagonals of a parallelogram bisect each other, using vectors

ANSWER Both midpoints are $\frac{1}{2}(OA + OC)$

4. A rotation of 90° anticlockwise about O sends $(3, 1)$ to:

ANSWER $(-1, 3)$

5. A reflection in $y = x$ sends $(4, -2)$ to:

ANSWER $(-2, 4)$

6. An enlargement centre $(1, 1)$, factor 3 sends $(3, 2)$ to:

ANSWER $(7, 4)$

7. Show that $ABCD$ with $A(0,0)$, $B(4,1)$, $C(6,5)$, $D(2,4)$ is a parallelogram

ANSWER $AB = DC = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$

07 Homework

Homework — 5 tasks

Every transformation answer must be a full description, not a name.

- With $A(-1, 3)$ and $B(5, -5)$, find AB , $|AB|$ and the unit vector along AB .
- Show that $P(2, 1)$, $Q(5, 7)$ and $R(9, 15)$ are collinear, and write the concluding sentence in full.
- Find k so that $\begin{pmatrix} 6 \\ k \end{pmatrix}$ is parallel to $\begin{pmatrix} 4 \\ 10 \end{pmatrix}$.

4. Describe fully the transformation that sends $(1, 2)$ to $(-2, 1)$ and $(3, 2)$ to $(-2, 3)$.
5. A triangle of area 9 is enlarged by factor $\frac{2}{3}$ about the origin. Find the area of the image.